

Electric Circuits

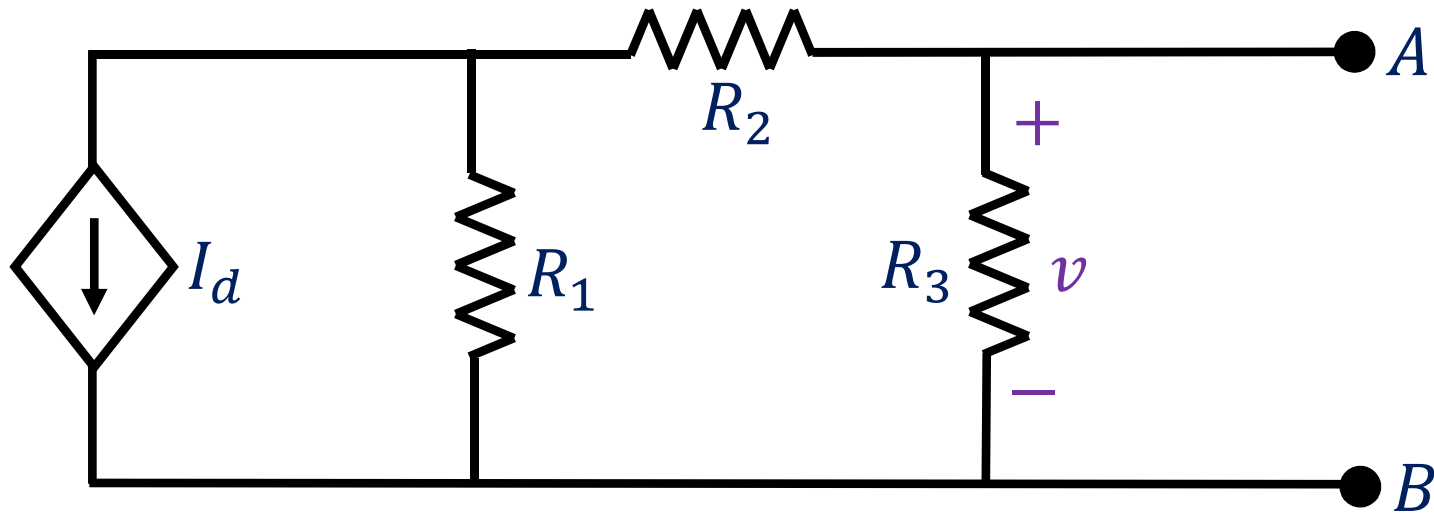


*Network Theorems: Maximum
Power Transfer Theorem*

Tentative Lecture Plan

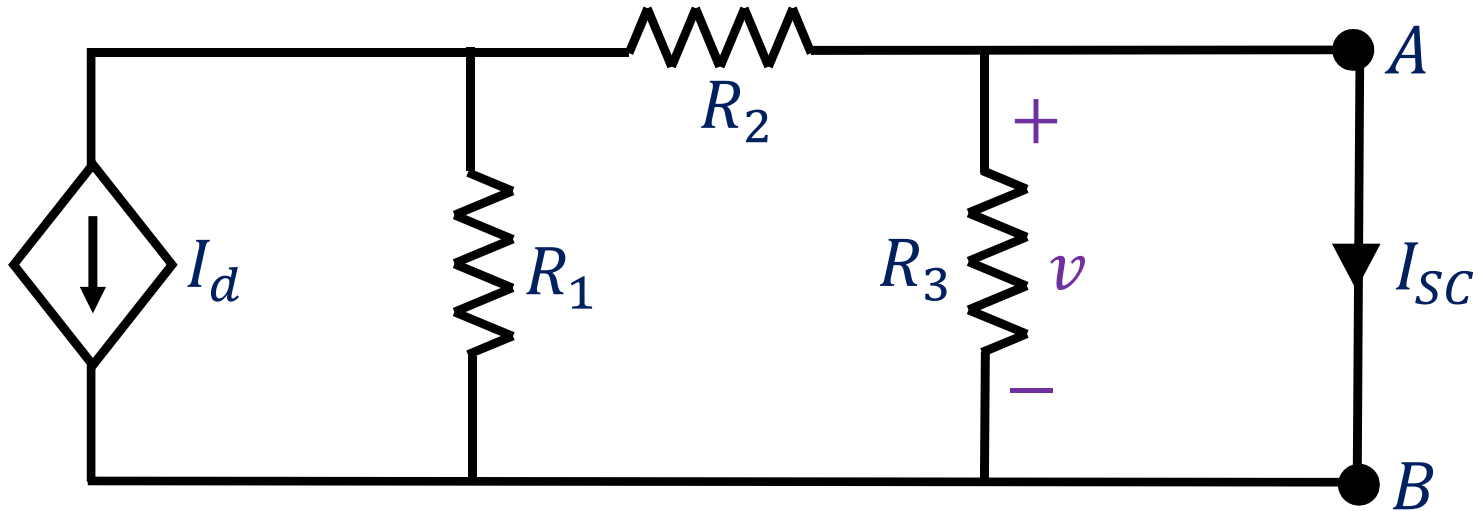
L	LDay-1	LDay-2	Topics	T	TDay
1	FRI, 04 AUG	MON, 07 AUG	Basic Definitions, Voltage & Current Sources, KCL & KVL	1	THU, 10 AUG
2	TUE, 08 AUG	FRI, 11 AUG	Nodal Analysis: Independent and Dependent Current Sources	2	SAT, 19 AUG
3	MON, 14 AUG	THU, 17 AUG	Nodal Analysis: Current and Voltage Sources		
4	FRI, 18 AUG	MON, 21 AUG	Mesh Analysis: Voltage and Current Sources	3	THU, 24 AUG
5	TUE, 22 AUG	FRI, 25 AUG	Network Theorems: Superposition Theorem	R	SAT, 26 AUG
SUN, 27 AUG : Quiz-1 : Lectures 1-5					
6	MON, 28 AUG	TUE, 29 AUG	Network Theorems: Thevenin's and Norton's Theorem	4	THU, 31 AUG
7	MON, 04 SEP	TUE, 05 SEP	Network Theorems: Maximum Power Transfer Theorem	5	WED, 06 SEP
8	FRI, 08 SEP	SAT (Mo), 09 SEP	Series RL and RC Circuits		
9	MON, 11 SEP	TUE, 12 SEP	Series RL, RC, RLC Circuits Response to Constant, Sinusoid and Exponential Input	6	THU, 14 SEP
10	FRI, 15 SEP	SAT (Tu), 16 SEP	Concept of Phasor, AC Circuits, AC Power		
WED, 20 SEP: MidSem : Lectures 1 – 10					

Draw Norton's Equivalent Network



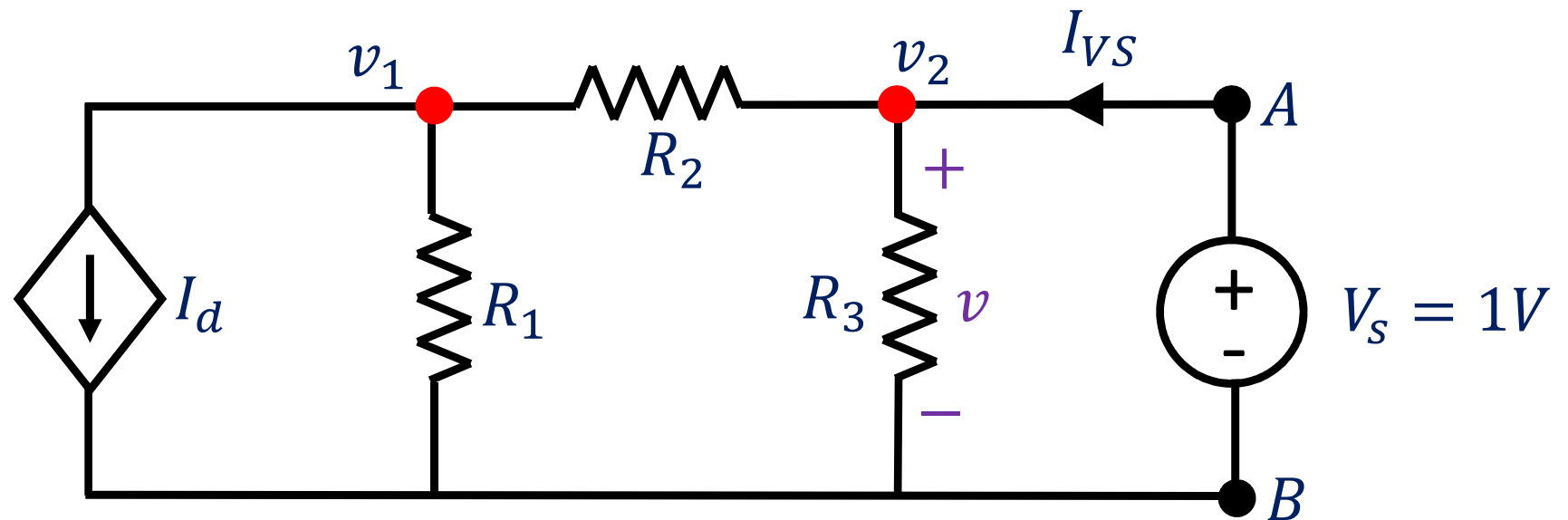
$$R_1 = 200\Omega \quad R_2 = 50\Omega \quad R_3 = 100\Omega \quad I_d = kv = 0.1v$$

Draw Norton's Equivalent Network

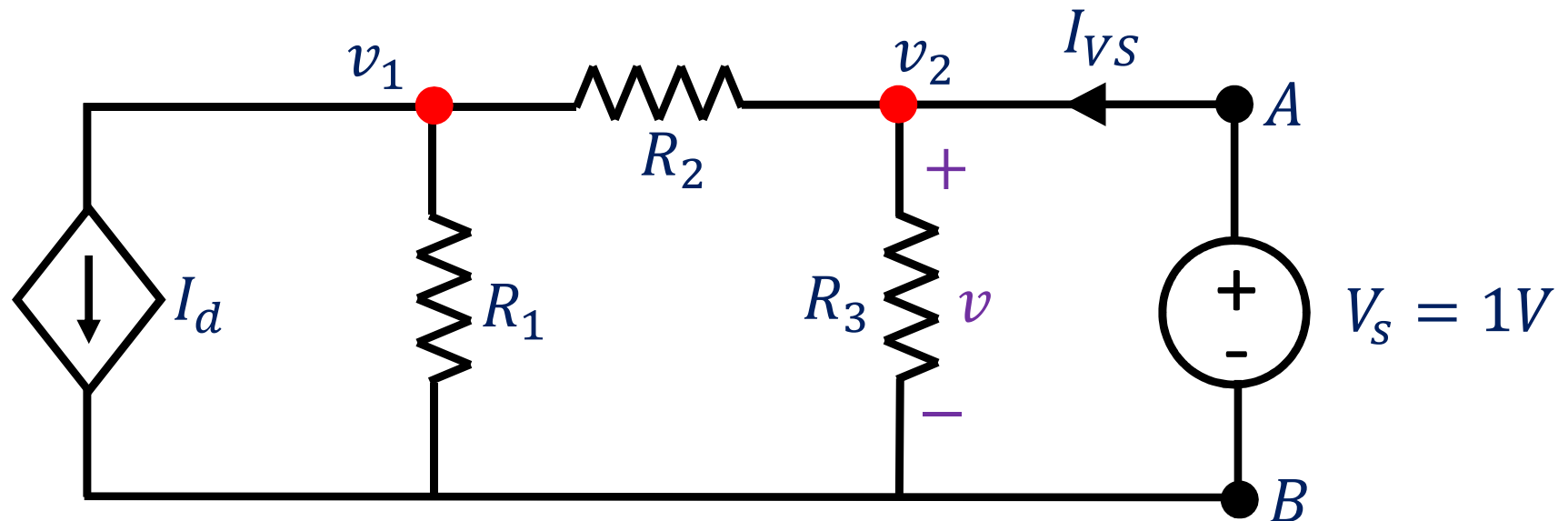


$$v = 0 \Rightarrow I_d = 0 \Rightarrow I_{SC} = 0$$

Draw Norton's Equivalent Network



Draw Norton's Equivalent Network



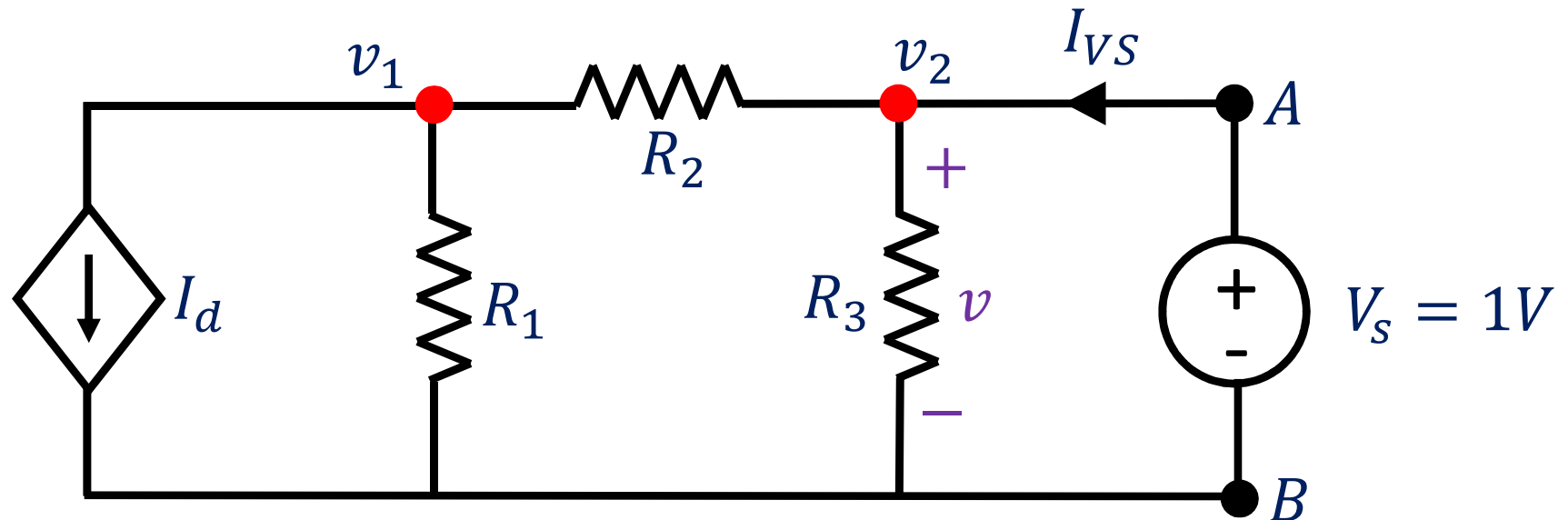
KCL at Node 1: $-I_d - G_1 v_1 - G_2(v_1 - v_2) = 0$

Rearranging: $v_1 = \left(\frac{G_2 - k}{G_1 + G_2} \right) V_s = -3.2V$

$$I_d = kv$$

$$v_2 = V_s$$

Draw Norton's Equivalent Network

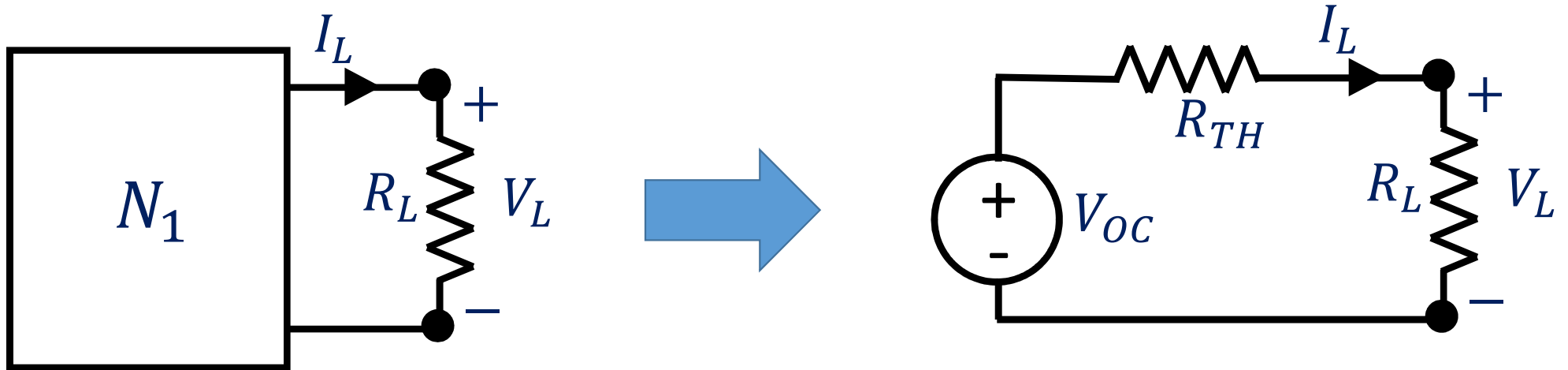


KCL at Node 2: $I_{VS} = G_2(v_2 - v_1) + G_3v_2 = 0.094A$ $v_2 = V_s$

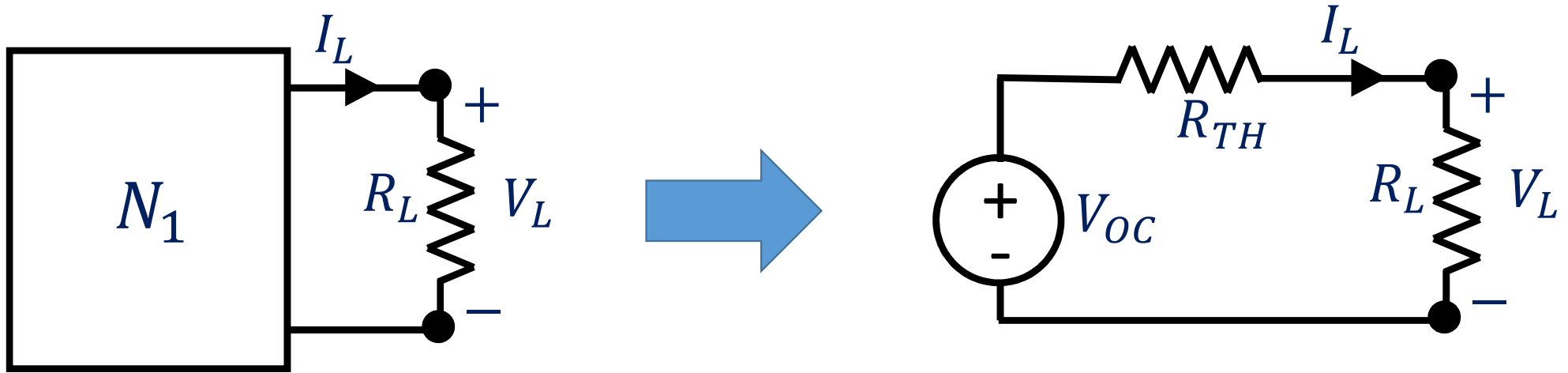
$$R_N = \frac{V_s}{I_{VS}} = \frac{1}{0.094} \Omega = 10.638\Omega$$

Maximum Power Transfer Theorem

Consider a linear network N_1 with its Thevenin's equivalent circuit as an independent voltage source V_{OC} in series with a resistance R_{TH} . The maximum power can be drawn from N_1 when the load resistance R_L equals R_{TH} . The maximum power $P_L^{max}(N_1)$ drawn from N_1 is given by $P_L^{max}(N_1) = \frac{V_{OC}^2}{4R_{TH}}$.



Maximum Power Transfer Theorem



Power P_L drawn by the Load R_L

$$P_L = \frac{V_{OC}^2 R_L}{(R_L + R_{TH})^2}$$

Maximum Power Transfer Theorem

$$P_L = \frac{V_{OC}^2 R_L}{(R_L + R_{TH})^2}$$

$$\frac{dP_L}{dR_L} = V_{OC}^2 \left\{ \frac{(R_L + R_{TH})^2 - 2R_L(R_L + R_{TH})}{(R_L + R_{TH})^4} \right\}$$

$$\frac{dP_L}{dR_L} = 0 \quad \Rightarrow \quad R_L = R_{TH} \quad \Rightarrow \quad P_L^{max} = \frac{V_{OC}^2}{4R_{TH}}$$

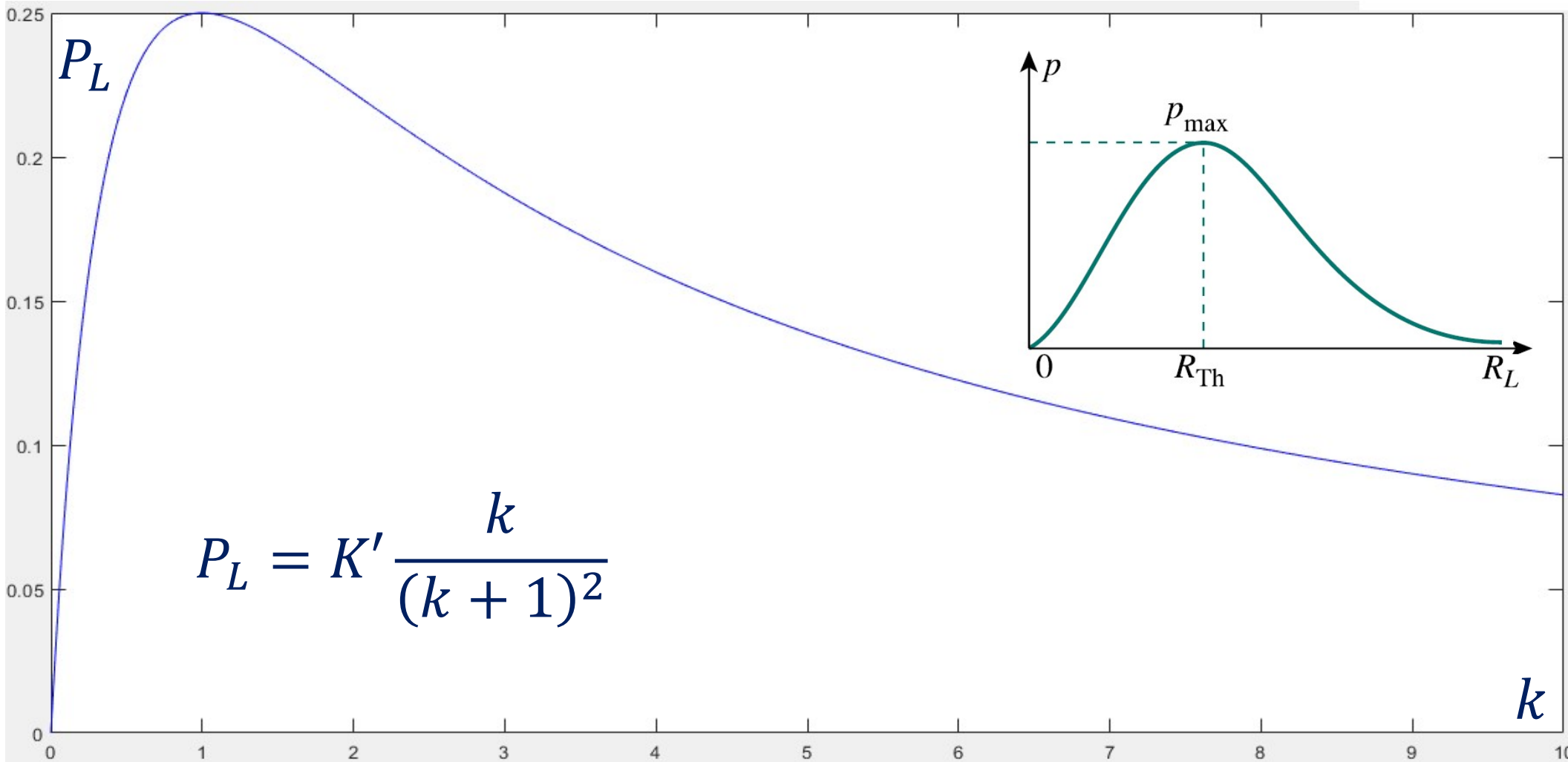
Power Delivered to Load

$$P_L = \frac{V_{OC}^2 R_L}{(R_L + R_{TH})^2}$$

$$P_L = V_{OC}^2 \left\{ \frac{\frac{R_L}{R_{TH}^2}}{\frac{(R_L + R_{TH})^2}{R_{TH}^2}} \right\} = \frac{V_{OC}^2}{R_{TH}} \frac{\left(\frac{R_L}{R_{TH}} \right)}{\left\{ \left(\frac{R_L}{R_{TH}} \right) + 1 \right\}^2}$$

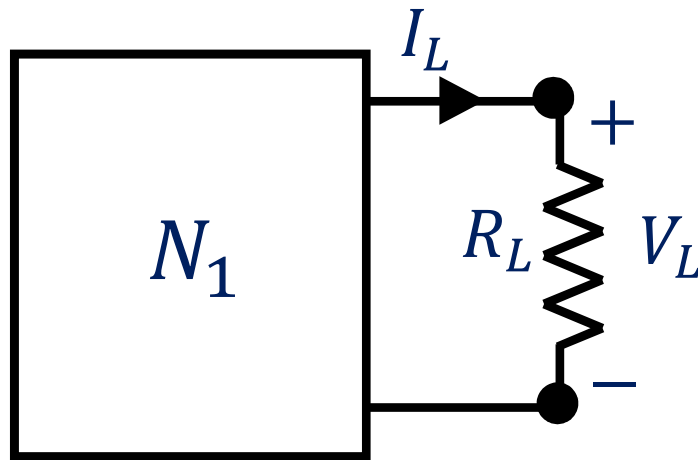
$$P_L = \frac{V_{OC}^2}{R_{TH}} \frac{k}{(k + 1)^2} \quad \longrightarrow \quad R_L = k R_{TH}$$

Power Delivered to Load



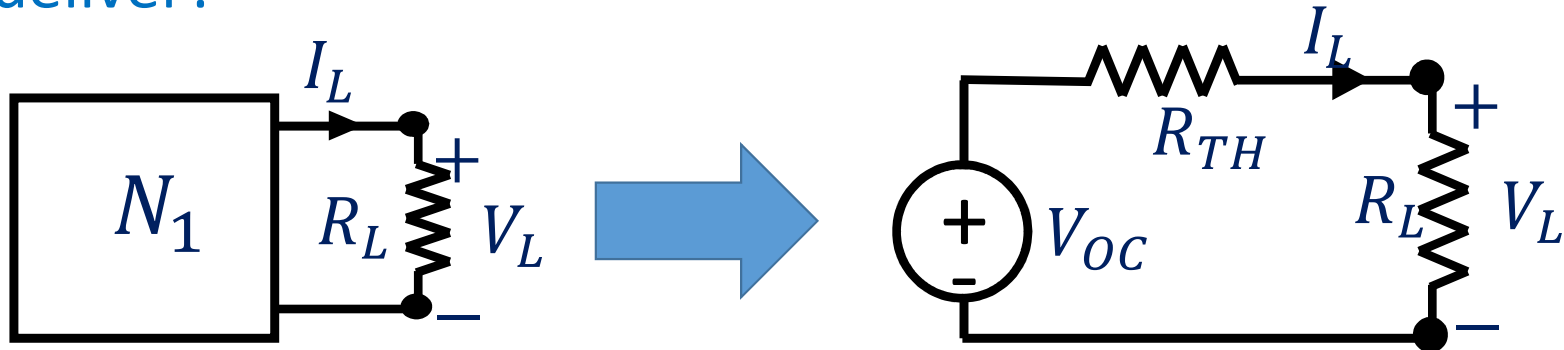
Maximum Power Transfer Theorem

A linear network N_1 delivers $4A$ when the load connected is 5Ω and $2A$ when the load is 20Ω . What is the maximum power that N_1 can deliver? Estimate its power transfer efficiency with $R_L = 5\Omega$. Compute the power transfer efficiency when $50W$ is delivered to the load.



Maximum Power Transfer Theorem

A linear network N_1 delivers $4A$ when the load connected is 5Ω and $2A$ when the load is 20Ω . What is the maximum power that N_1 can deliver?



$$\left. \begin{aligned} V_{OC} &= 4(R_{TH} + 5) \\ V_{OC} &= 2(R_{TH} + 20) \end{aligned} \right\} \begin{aligned} R_{TH} &= 10\Omega \\ V_{OC} &= 60V \end{aligned} \rightarrow P_L^{max}(N_1) = 90W$$

Maximum Power Transfer Theorem

Estimate its power transfer efficiency with $R_L = 5\Omega$. Compute the power transfer efficiency when $50W$ is delivered to the load.

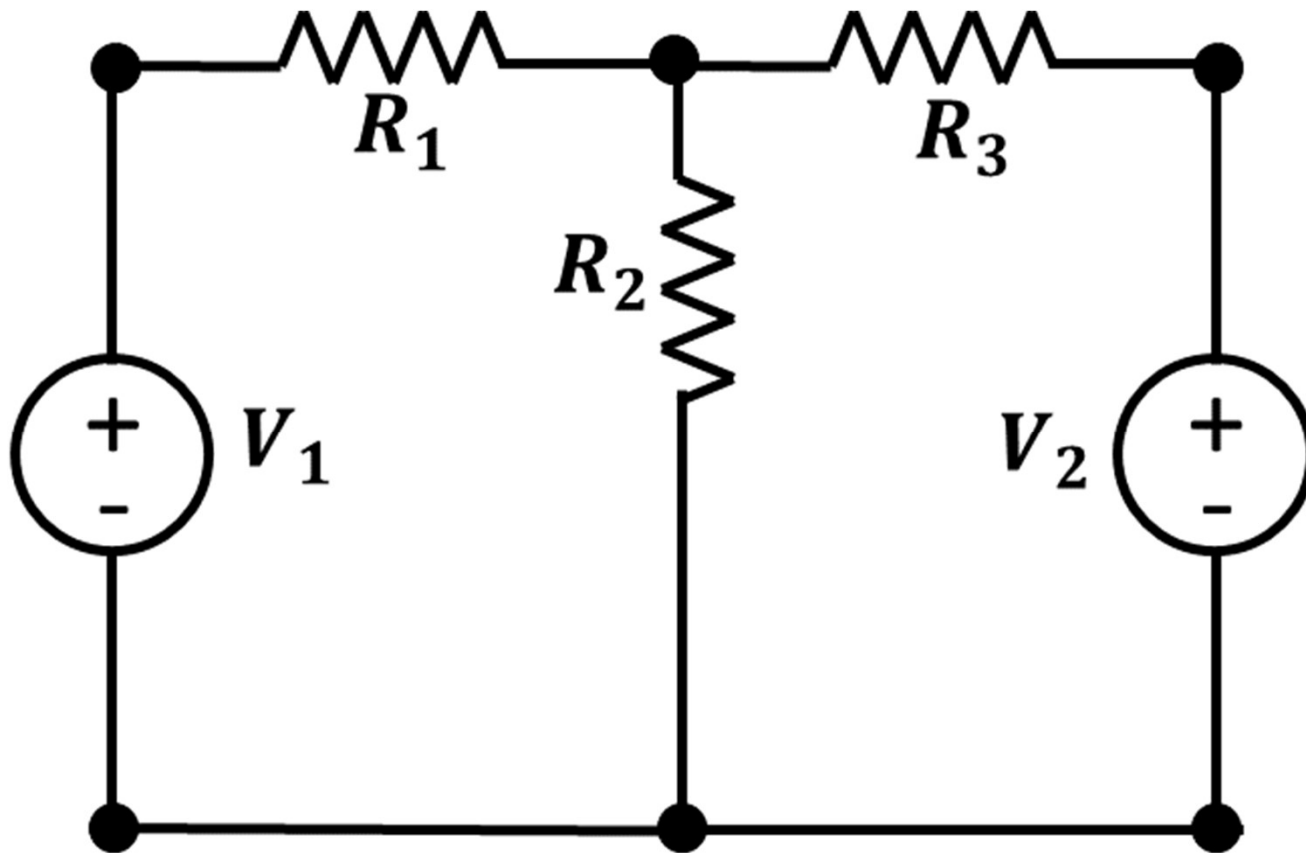
$$\eta_P(R_L = 5) = \frac{R_L}{R_L + R_{TH}} = \frac{5}{5 + 10} = 33.33\%$$

$$V_{OC} = 60V$$
$$R_{TH} = 10\Omega$$

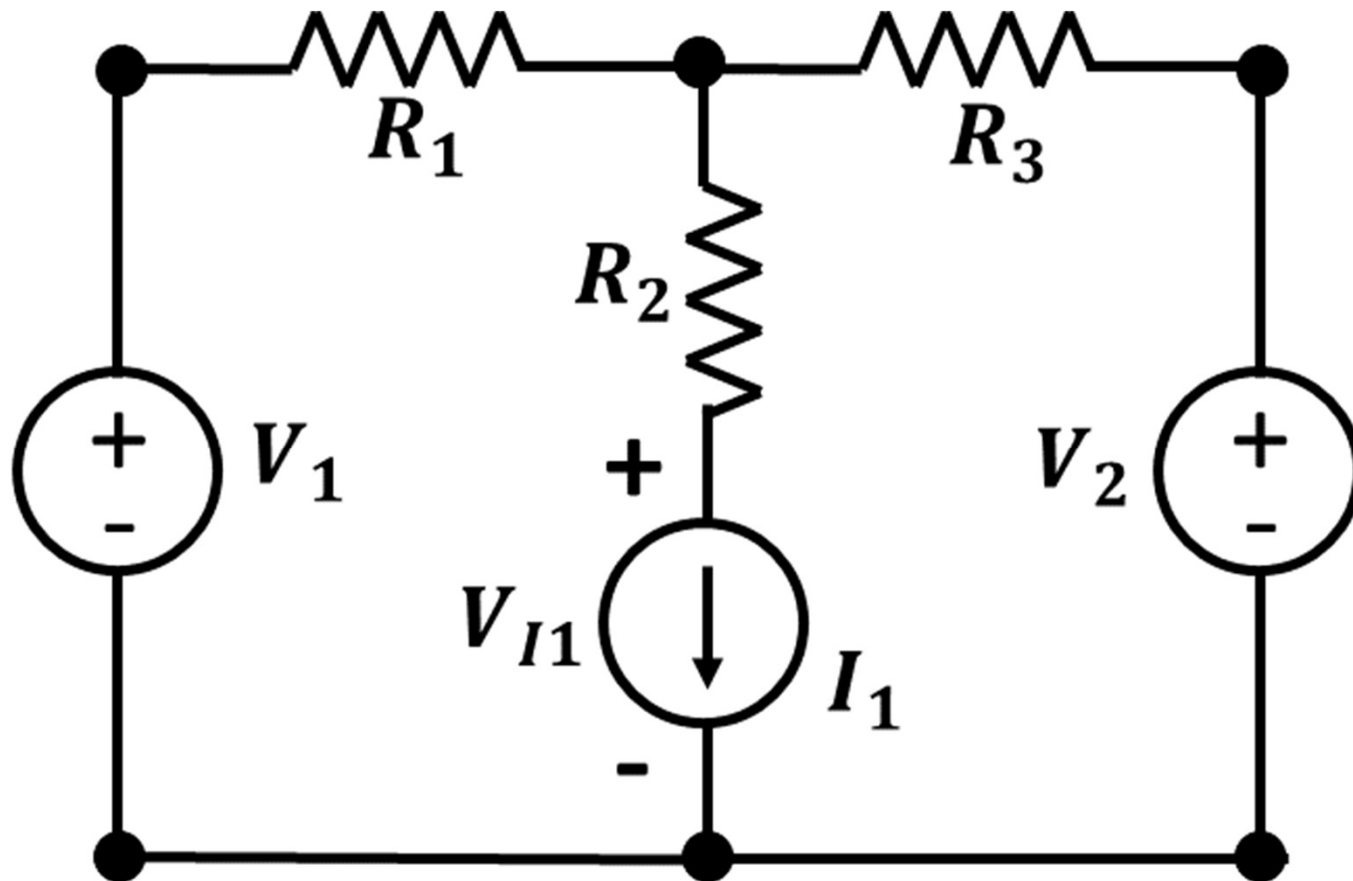
$$P_L(N_1) = 50W \quad \longrightarrow \quad 50 = \frac{V_{OC}^2 R_L}{(R_L + R_{TH})^2} = \frac{3600 R_L}{(R_L + 10)^2}$$

$$50R_L^2 - 2600R_L + 5000 = 0 \quad \begin{cases} \nearrow R_L = 50\Omega \\ \searrow R_L = 2\Omega \end{cases}$$
$$\eta_P(R_L = 50) = 83.33\%$$
$$\eta_P(R_L = 2) = 16.67\%$$

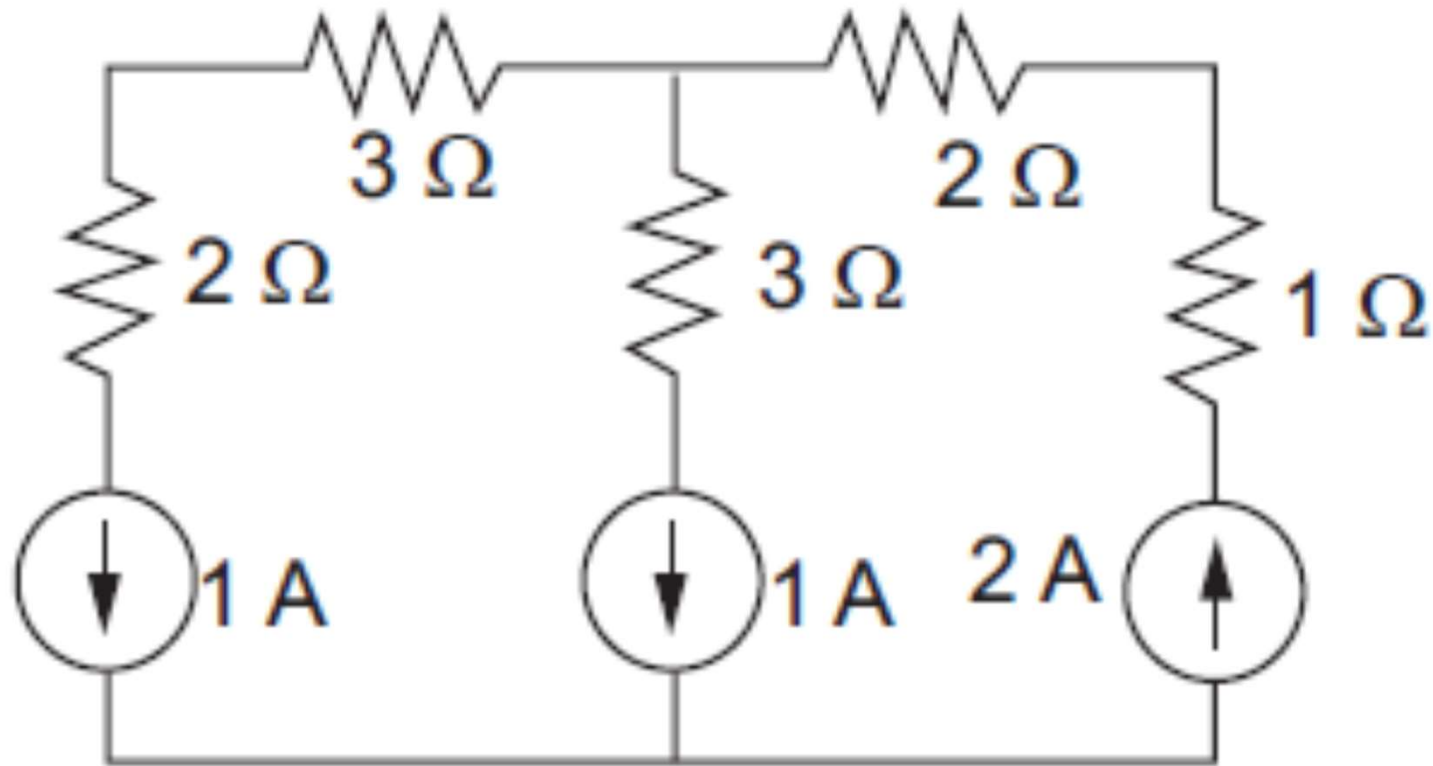
Solve for the Branch Voltages and Currents



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Solve for the Branch Voltages and Currents



Lumped and Distributed Elements

An element is classified as a *lumped element* if the net effect of the electrical phenomena taking place within that element can be described in terms of only its terminal voltage and current variables, irrespective of its internal details and geometry.

If the electrical description of an element calls for voltage and current variables that are functions of space variables over the element, then it is called a *distributed element*.

Linear and Time Invariant Elements

An element is classified as a *linear element* if its $v - i$ relationship satisfies the principles of superposition .

$$v(t) = Ri(t)$$

$$v(t) = L \frac{di(t)}{dt}$$

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(t) dt$$

An element is classified as *time invariant* if the parameters that characterize it are independent of time.

$$R(t) = R$$

Summary

- Solving Networks using KCL & KVL
 - Variables: $2b = (n - 1) + \{b - (n - 1)\}$
- Solution using Nodal Analysis
 - Node Voltage Variables: $n - 1$
- Solution using Mesh Analysis
 - Mesh Current Variables: $b - n + 1$
- Superposition Theorem: Linearity Property of Circuits
- Thevenin's and Norton's Equivalent Networks
- Maximum Power Transfer Network



Thank You